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"99.9% of science is boring."
- Makise Kurisu

PRECALCULUS

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CONTENTS

1 Functions

1.1 Intro To Functions

Definition 1.1 (Function). A function is a relationsip that maps 1 input to 1 output.

Note: An input never gives more than 1 output.

A function usually looks like

$$y = f(x)$$

x is the *independent* variable.

y or $f(x)$ is the *Dependent* variable.

Where the standard ordered pairs looks like

$$(x, y) \vee (x, f(x))$$

Definition 1.2 (Domain). A domain is a set of inputs.

Definition 1.3 (Range). A range is a set of outputs.

Example 1.1 (Real Example).

Work Hours	Pay
25	\$500
53	\$310
30	\$490
40	\$490

Note: This is a function but not a *one-to-one function*.

$\mathrm{Domain}: 23, 53, 30, 40$
$\mathrm{Range}: \$500, \$310, \$490$

Example 1.2 (Input Output).

$$S = (-2, 16), (-1, 4), (0, 3), (1, 4)$$

This is a function.

$$R = (-2, 5), (3, 9), (5, 0), (-2, 6)$$

This is not a function.

Example 1.3 (Algebraic).

$$y = \frac{1}{2}x - 2$$

This is a function.

$$y = 2x^2 - 5x + 4$$

This is a function.

$$y = \pm\sqrt{3-2x}$$

This is not a function because we get more than 1 output for only 1 input.

Note: $y^2 = x \neq y = \sqrt{x}$, instead it equals to $y = \pm\sqrt{x}$. Which makes it not a function.

Example 1.4 (Factoring).

$$xy + 2y = 3x - 1$$

$$y(x+2) = 3x - 1$$

$$y = \frac{3x-1}{x+2}$$

1.2 Evaluating Functions

Note:

$$y = -7x + 5 \equiv f(x) - 7x + 5$$
$$y = \frac{3}{2}x - 1 \equiv g(x) = \frac{3}{2}x - 1$$

We use the $f(x)$ notation to know what the input is.

Example 1.5 (Exponents).

$$h(x) = -2x^2 + x - 1$$

1. $h(-1) = -2(-1)^2 + (-1) - 1$
   $h(-1) = -4$
   $(-1, -4)$
2. $h(-x) = -2(-x)^2 + (-x) - 1$
   $h(-x) = -2x^2 + x - 1$
3. $h(x+1) = -2(x+1)^2 + (x+1) - 1$
   $h(x+1) = -2(x^2 + 2x + 1) + (x+1) - 1$
   $h(x+1) = -2x^2 - 4x - 2 + x + 1 - 1$
   $h(x+1) = -2x^2 - 3x - 1$

Example 1.6 (Square Root).

$$f(x) = \sqrt{x^2 - 3x}$$

1. $f(5) = \sqrt{(5)^2 - 3(5)}$
   $f(5) = \sqrt{10}$
   $(5, \sqrt{10})$
2. $f(0) = \sqrt{(0)^2 - 3(0)}$
   $f(0) = \sqrt{0}$
   $(0, 0)$
3. $f(2x) = \sqrt{(2x)^2 - 3(2x)}$
   $f(2x) = \sqrt{4x^2 - 6x}$
4. $f(x+h) = \sqrt{(x+h)^2 - 3(x+h)}$
   $f(x+h) = \sqrt{x^2 + 2xh + h^2 - 3x - 3h}$
5. $f(1) = \sqrt{(1)^2 - 3(1)}$
   $f(1) = \sqrt{-2}$
   Note: This is a domain issue so we can't use 1.

Note: For domains:

1. **Square Roots:** The inside of the square root *(radicand)* must always always positive.
2. **Denominator:** The denominator must never be 0.

Example 1.7 (Fractions).

$$g(x) = \frac{2x + 1}{x - 5}$$

1. $g(0) = \frac{2(0)+1}{(0)-5}$
   $g(0) = \frac{1}{-5}$

2. $g(x^2) = \frac{2(x^2)+1}{(x^2)-5}$
   $g(x^2) = \frac{2x^2+1}{x^2-5}$

3. $g(5) = \frac{2(5)+1}{(5)-5}$
   $g(5) = \frac{11}{0}$
   Note: This is also a domain issue.

4. $-g(x) = -\frac{2x+1}{x-5}$

1.2.1 Difference Quotient

Definition 1.4 (Difference Quotient). A formula used in algebra and calculus to find the **average rate of change** of a function over a small interval

$$\frac{f(x+h)-f(x)}{h}$$

Example 1.8 (Example).

$$f(x) = x^2 + 1$$

$f(x+h) = (x+h)^2 + 1$
$f(x+h) = x^2 + xh + xh + h^2 + 1$
$f(x+h) = x^2 + 2xh + h^2 + 1$

$\frac{f(x+h)-f(x)}{h} = \frac{(x^2+2xh+h^2+1)-(x^2+1)}{h}$
$\frac{f(x+h)-f(x)}{h} = \frac{2xh+h^2}{h}$
$\frac{f(x+h)-f(x)}{h} = \frac{h(2x+h)}{h}$
$\frac{f(x+h)-f(x)}{h} = 2x + h$

1.3 Finding Domains

Definition 1.5 (Domains). Domains is the set of inputs of a function that give a $\mathbb{R}$ number output.

1.3.1 The Problem Areas

1. **Square Roots:** $\sqrt{()}$ radicand > 0 .
2. **Denominators:** $\frac{}{()} \neq$ zero.

Example 1.9 (Square Root).

$$f(x) = \sqrt{5 - 4x}$$
$$5 - 4x \geq 0$$
$$-4x \geq -5$$
$$x \geq \frac{-5}{-4}$$
$$x \leq \frac{5}{4}$$

Can be written as

$$D = \{x \mid x \leq \frac{5}{4}\}$$

Or

$$(-\infty, \frac{5}{4}]$$

Example 1.10 (Denominator).

$$g(t) = \frac{5t}{t^3 - 16t}$$
$$t^3 - 16t = 0$$
$$t(t^2 - 16) = 0$$
$$t(t-4)(t+4)$$
$$t \neq 0 \vee t \neq 4 \vee t \neq -4$$

Can be written as

$$D = \{t \in \mathbb{R} \mid t \neq 0, t \neq -4, t \neq 4\}$$

Or

$$(-\infty, -4) \cup (-4, 0) \cup (0, 4) \cup (4, \infty)$$

Example 1.11 (Examples).

1. $f(x) = 3x^2 - 2x + 4$
   $D : \{x \in \mathbb{R}\}$ Or $(-\infty, \infty)$

2. $h(x) = \frac{\sqrt{3x-15}}{x-4}$
   $x \neq 4$
   $3x - 15 \geq 0$
   $3x \geq 15$
   $x \geq 5$
   $D : \{x \mid x \neq 5\}$ or $[5, \infty)$
3. $g(x) = \frac{5}{\sqrt{x-8}}$
   $x - 8 > 0$
   $D : \{x \mid x > 8\}$ or (x, ∞)

1.4 Operations Of Functions

Definition 1.6 (Operations Of Function). Function operations invole combining two or more functions to create a new function using standard arithmetic: **addition, subtraction, multiplication** and **division**.

Example 1.12 (Operations).

$$f(x) = \frac{2x+3}{3x-2} \qquad g(x) = \frac{4x}{3x-2}$$

$$D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

1. $$(f+g)(x) = \frac{2x+3}{3x-2} + \frac{4x}{3x-2} \to \frac{6x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

2. $$(f-g)(x) = \frac{2x+3}{3x-2} - \frac{4x}{3x-2} \to \frac{-2x+3}{3x-2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

3. $$(f \cdot g)(x) = \left(\frac{2x+3}{3x-2}\right)\left(\frac{4x}{3x-2}\right) \to \frac{8x^2+12x}{(3x-2)(3x-2)} \to \frac{8x^2+12x}{(3x-2)^2} \quad D : \left\{ x \mid x \neq \frac{2}{3} \right\}$$

4. $$\left(\frac{f}{g}\right)(x) = \frac{\frac{2x+3}{3x-2}}{\frac{4x}{3x-2}} \to \frac{2x+3}{3x-2} \cdot \frac{3x-2}{4x} \to \frac{2x+3}{4x} \quad D \left\{ x \mid x \neq \frac{2}{3}, x \neq 0 \right\}$$

Example 1.13 (Square Root and Fractions).

$$f(x) = \sqrt{x+3} \qquad g(x) = \frac{5}{x}$$

$$D : \{x \mid \geq -3\} \qquad D : \{x \mid x \neq 0\}$$

1. $$(f+g)(x) = \sqrt{x+3} + \frac{5}{x} \qquad D : \{x \mid x \geq -3, x \neq 0\}$$

2. $$(f-g)(x) = \sqrt{x+3} - \frac{5}{x} \qquad D : \{x \mid x \geq -3, x \neq 0\}$$

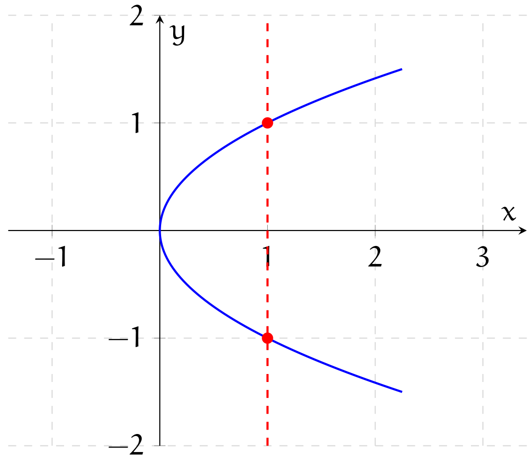
3. $$(f \cdot g)(x) = \frac{5\sqrt{x+3}}{x} \qquad D : \{x \mid x \geq -3, x \neq 0\}$$

4. $$\left(\frac{f}{g}\right)(x) = \sqrt{x+3} \cdot \frac{x}{5} \to \frac{x\sqrt{x+3}}{5} \qquad D : \{x \mid x \geq -3, x \neq 0\}$$

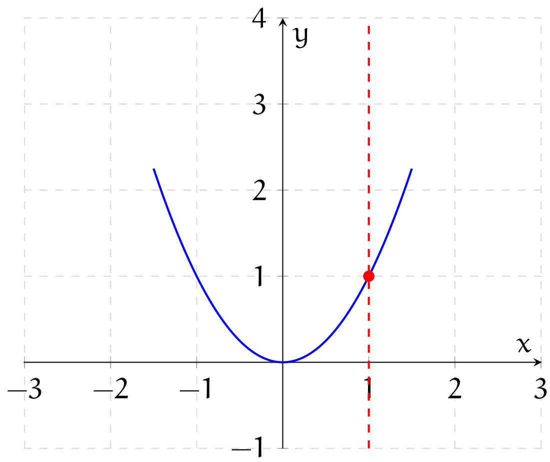
1.5 The Vertical Line Test

Definition 1.7 (Vertical Line Test). This is a test to determine if a graph is a function or not.

Example 1.14 (Non Function). This is not a function because the vertical line passes through 2 points, which means that it outputs more than 1 output.



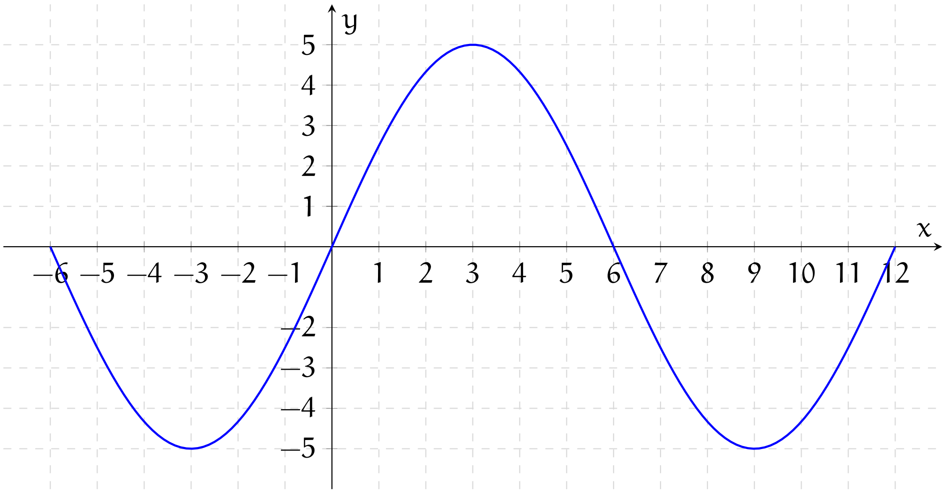
Example 1.15 (Function). This is a function because the vertical line passes through only 1 point, which means that it only gives a single output.



1.6 Features of Graphs Points, Domain, Range

Definition 1.8 (Graph Points). A graph point of a function is an ordered pair (x, y) on a coordinate plane, where y is the output generated when the function precesses the input x.

Example 1.16 (Graphs).



1. Find $f(0) = 0 \quad (0, 0)$
2. Find $f(-2) = -2 \quad (-2, -2)$
3. Is $f(1)\, positive \vee\, negative \;=\; positive$
4. Is $f(-2)\, positive \vee\, negative \;=\; negative$
5. For what x is $f(x) = 0$?
   $(-6, 0)\ (0, 0)\ (6, 0)\ (12, 0) \quad x = -6,\ 0,\ 6,\ 12$
6. For what x is $f(x) > 0$?
   $0 < x < 6 \quad (-4, 6)$
7. For what x is $f(x) < 0$?
   $-6 < x < 0,\ 6 < x < 12 \quad (-6, 0) \cup (6, 12)$
8. Domain: $D : \{x \mid -6 \le x \le 12\} \quad [-6, 12]$
9. Range: $R : \{f(x) \mid -5 \le f(x) \le 5\} \quad [-5, 5]$
10. x Intercept: $(-6, 0)$, $(0, 0)$, $(6, 0)$, $(12, 0)$
11. y intercept: $(0, 0)$
12. How many times does the $y = 2$ cross the graph? 2
13. For what x is $f(x) = 4$?
    $(2, 4)\ (4, 4)$
14. For what x is $f(x) = -5$?
    $(-3, -5)\ (9, -5)$

Definition 1.9 (Square Brackets: Inclusive). “[“ or ”]“ The endpoint number is included in the interval (corresponds to $\le \vee \ge$).

Definition 1.10 (Parentheses: Exclusive). "(" or ")" The endpoint number is not included in the interval (corresponds to $< \vee >$). It is also always used for infinity (∞).

Example 1.17 (Algebraic).

$$f(x) = \frac{x+2}{x-6}$$

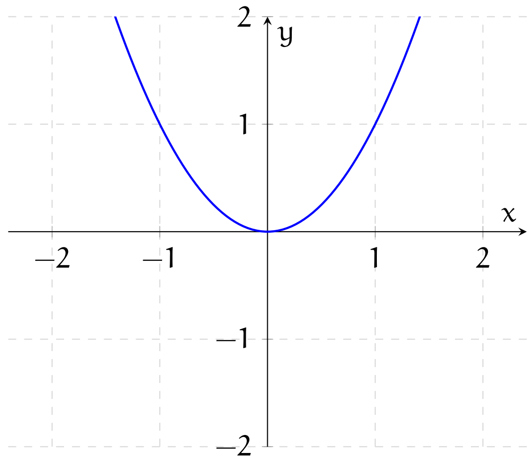
1. $f(1) = \frac{(1)+2}{(1)-6} \rightarrow \frac{3}{5} \quad \left(1, -\frac{3}{5}\right)$
2. $f(x) = 2$ ... find x
   $2 = \frac{x+2}{x-6}$
   $2x - 12 = x + 2$
   $x - 12 = 2$
   $x = 14$
   $(14, 2)$
3. Domain:
   $x - 6 = 0$
   $x = 6$
   $D = \{x \mid x \neq 6\}$
   $(-\infty, 6) \cup (6, \infty)$
4. x intercept: $f(x) = 0$
   $0 = \frac{x+2}{x-6}$
   $x + 2 = 0$
   $x = -2$
   $(-2, 0)$
5. y intercept:
   $f(0) = \frac{(0)+2}{(0)-6} = -\frac{1}{3}$
   $\left(0, -\frac{1}{3}\right)$

1.7 Properties of Functions: Even or Odd

Definition 1.11 (Even). Even functions are symmetric about y-axes in a graph.

Example 1.18 (Even Function). This is an even function because if we fold it down the y-axis and fold it onto itself, it would be a mirror image perfectly.

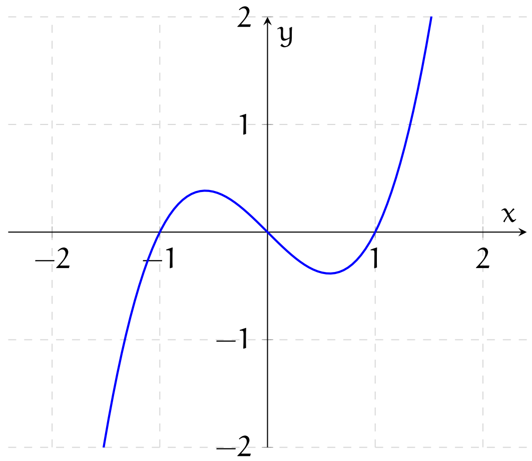
$$f(x) = x^2$$



Definition 1.12 (Odd). Odd functions are symmetric about origins in a graph.

Example 1.19 (Odd Function). This is an odd function because if we rotate the graph 180° about the origin, it will still be the exact same graph.

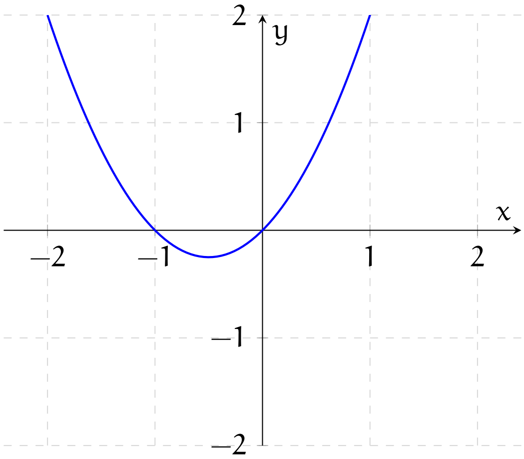
$$f(x) = x^3 - x$$



Note: A lot of graphs are neither even or odd. For example:

Example 1.20 (Neither).

$$f(x) = x^2 + x$$



Definition 1.13 (Algebraic Definition of Even Functions).

$$f(x) = f(-x)$$

Definition 1.14 (Algebraic Definition of Odd Functions).

$$-f(x) = f(-x)$$

Example 1.21 (Algebraic).

- $f(x) = 2x^4 - x^2$
  $f(-x) = 2(-x)^4 - (-x)^2$
  $f(-x) = 2x^4 - x^2 \quad \text{Even}$
- $g(x) = \frac{x}{x^2-1}$
  $g(-x) = \frac{(-x)^1}{(-x)^2-1} \to g(-x) = \frac{-x}{x^2-1}$
  $g(-x) = -\frac{x}{x^2-1} \quad \text{Odd}$
- $h(x) = 3x^3 + 5$
  $h(-x) = 3(-x)^3 + 5 \to h(-x) = -3x^3 + 5$
  $h(-x) = -(3x^3 - 5) \quad \text{Neither}$
- $F(x) = \frac{x^2+3}{x^2-1}$
  $F(-x) = \frac{(-x)^2+3}{(-x)^2-1}$
  $F(-x) = \frac{x^2+3}{x^2-1} \quad \text{Even}$
- $G(x) = \sqrt{x}$
  $G(-x) = \sqrt{(-x)} \quad \text{Neither because it's not a } \mathbb{R} \text{ number}$
- $H(x) = \frac{-x^3}{3x^2-9} \to -\frac{x^3}{3x^2-9}$
  $H(-x) = \frac{-(-x)^3}{3x^2-9}$
  $H(-x) = \frac{x^3}{3x^2-9} \quad \text{Odd}$

1.8 Properties of Functions: Increasing or Decreasing

Definition 1.15 (Increasing).

$$\text{for } x_1 < x_2$$

$$f(x_1) < f(x_2)$$

From $\text{left} \to \text{right}$ in the x-axis, the graph climbs upward.

Definition 1.16 (Decreasing).

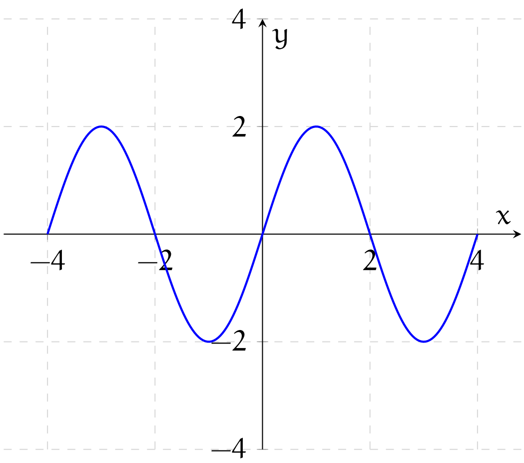
$$\text{for } x_1 < x_2$$

$$f(x_1) > f(x_2)$$

From $\text{left} \to \text{right}$ in the x-axis, the graph falls downward.

Note: All of this is given by an open intervals of the x-axes.

Example 1.22 (Decreasing and Increasing Function).

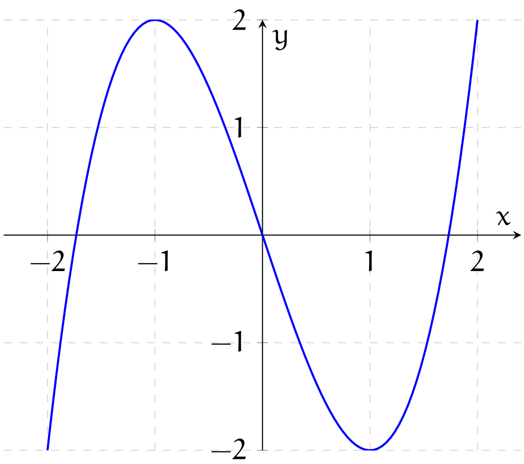


$$\text{Increasing } [-1, 1] \cup [3, 4]$$

$$\text{Decreasing } [-3, -1] \cup [1, 3]$$

Another example is:

$$f(x) = -x^3 - 3 * x$$



$$\text{Increasing } [-1, 1]$$

$$\text{Decreasing } (-\infty, -1] \cup [1, \infty)$$

1.9 Properties of Functions: Extremas

Definition 1.17 (Local/Relative Maximum). The highest point in a specific region or a small interval of a function.

$$f(x) \leq f(c) \ldots c \text{ is a local max}$$

Where functions changes from increasing to decreasing.

Definition 1.18 (Local/Relative Minimum). The lowest point in a specifis region or a small interval of a function.

$$f(x) \geq f(c) \ldots c \text{ is a local min}$$

Where functions changes from decreasing to increasing.

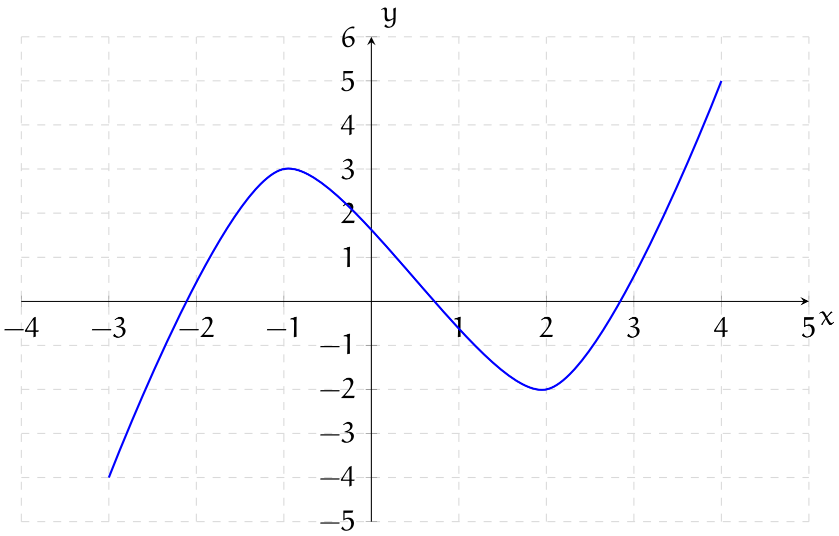
Note: Endpoints are not allowed for Local minimum or maximum. But it is allowed for absolute minimum or maximum.

Definition 1.19 (Absolute Maximum). Highest output on graph.

Definition 1.20 (Absolute Minimum). Lowest output on graph

Note: Every continous closed interval "[,]" will guarantee we have an absolute max and an absolute min.

Example 1.23 (Graph).



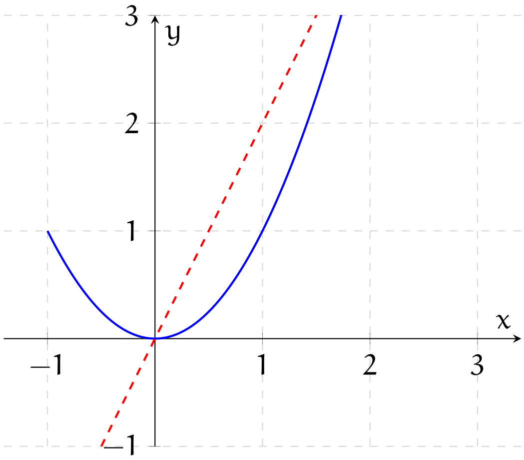
1. Local Minimum: $-2 @ (2, -2)$
2. Local Maximum: $3 @ (-1, 3)$
3. Absolute Minimum: $-4 @ (-3, -4)$
4. Absolute Maximum: $5 @ (4, 5)$

1.10 Average Rate of Change

Definition 1.21 (Average Rate of Change). The slope of the line connecting two points on a curve, known as the *secant line*.

$$\text{Average Rate of Change } = \frac{f(b) - f(a)}{b - a}$$

Example 1.24 (Graph).



$$a = 0, b = 2$$
$$\frac{f(2) - f(0)}{2 - 0} = \frac{4 - 0}{2 - 0} = 2$$

Example 1.25 (Algebraic).

$$f(x) = x^2 - 2x \text{ Find average rate of change}$$

$$\text{From } x = 3 \text{ to } x = 5$$
$$f(3) = (3)^2 - 2(3)$$
$$f(3) = 3$$
$$f(5) = (5)^2 - 2(5)$$
$$f(5) = 15$$
$$(3, 3)\ (5, 15)$$
$$\text{Average Rate of Change } = \frac{15 - 3}{5 - 3}$$
$$\text{Average Rate of Change } = \frac{12}{2} \to 6$$

1.11 Piece-wise Functions

Definition 1.22 (Piece-wise Functions). A single function that uses two or more different formulas or rules, with each rule applying to a specific portion of the function's domain.

Example 1.26 (Algebraic).

1.
$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$

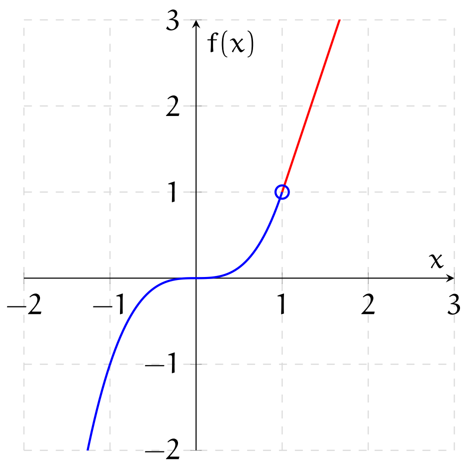
   (a) $f(-1) = (-1)^3 \rightarrow (-1, -1)$
   (b) $f(0) = (0)^3 \rightarrow (0, 0)$
   (c) $f(1) = 3(1) - 2 \rightarrow (1, 1)$

2.
$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

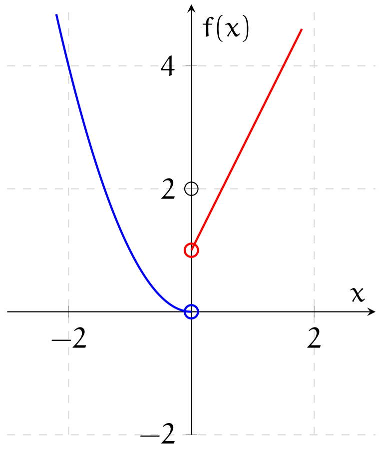
   (a) $f(-2) = (-2)^2 \rightarrow (-2, 4)$
   (b) $f(0) = 2 \rightarrow (0, 2)$
   (c) $f(2) = 2(2) + 1 \rightarrow (2, 5)$

Example 1.27 (Graph).

1.
$$f(x) = \begin{cases} x^3 & \text{if } x < 1 \\ 3x - 2 & \text{if } x \geq 1 \end{cases}$$



2.
$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ 2x + 1 & \text{if } x > 0 \end{cases}$$

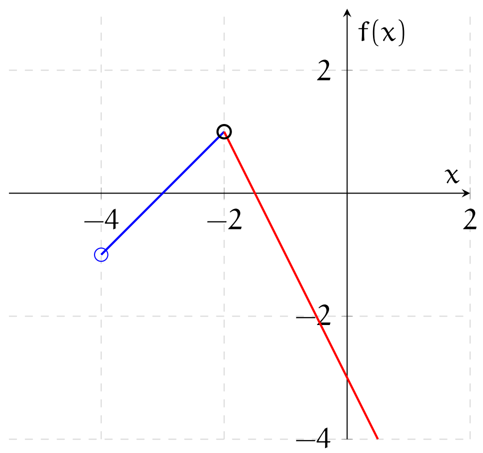


Definition 1.23 (Not Continuous Function). In a piece-wise function, a function is not continuous *(discontinuous)* at a point if the different sub-functions do not connect seamlessly at the transition boundary

Example 1.28 (Not Continouous Function).

$$f(x) = \begin{cases} x+3 & \text{if } -4 \leq x < -2 \\ -2x-3 & \text{if } x > -2 \end{cases}$$

Example 1.29 (Graph).

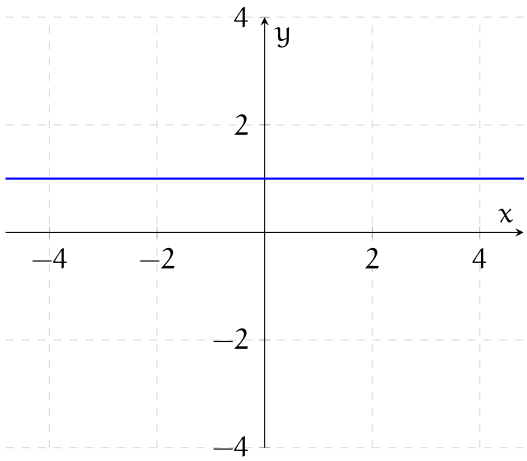


1.12 Important Functions

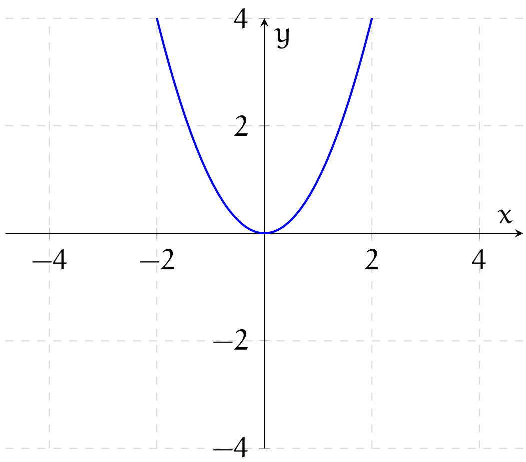
Example 1.30 (Even). All of the examples have the point $(0,0)$ $(1,1)$ $(-1,1)$ except for the $f(x) = b$ which is just a constant value and the $f(x) = \sqrt{x}$ function which can't have a negative input ($x \geq 0$).

1.
$$f(x) = b$$

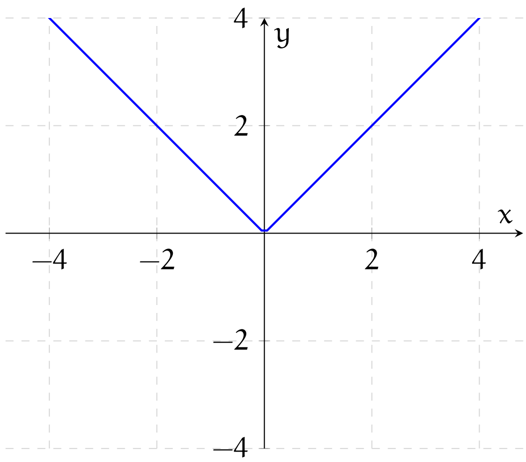
For this example $b = 1$



2.
$$f(x) = x^2$$

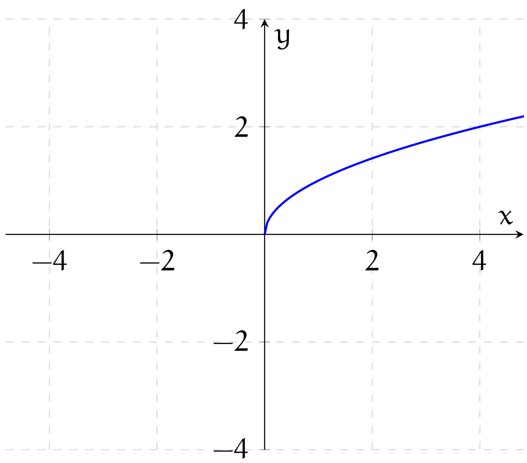


3.
$$f(x) = |x|$$



4.

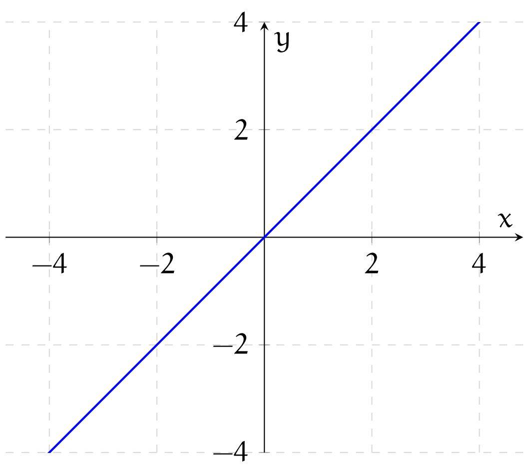
$$f(x) = \sqrt{x}$$



Example 1.31 (Odd). All of the examples have the point $(0, 0)$ $(1, 1)$ $(-1, -1)$ except for the $f(x) = \frac{1}{x}$ function which doesn't have the $(0, 0)$ point.

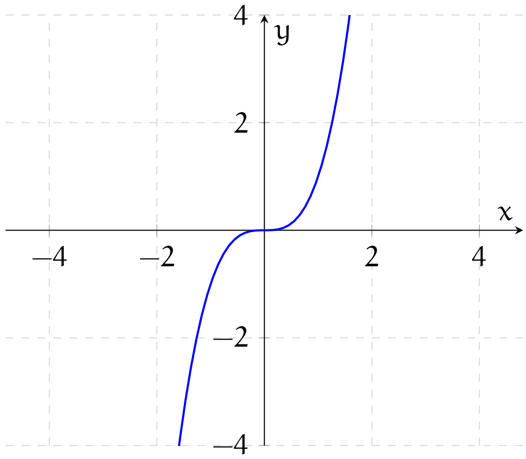
1.

$$f(x) = x$$



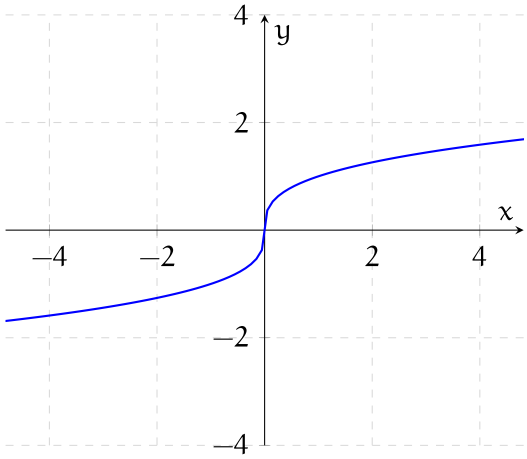
2.

$$f(x) = x^3$$



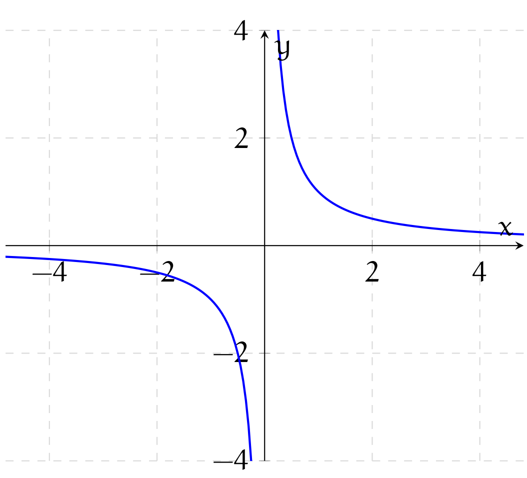
3.

$$f(x) = \sqrt[3]{x}$$



4.

$$f(x) = \frac{1}{x}$$



2 Graphical Transformations

2.1 Introduction To Graphical Transformations

Definition 2.1 (Graphical Transformations). A mathematical operation that alters the position, size or orientation of a function's graph on a coordinate plane, without changing its basic shape.

2.1.1 Vertical

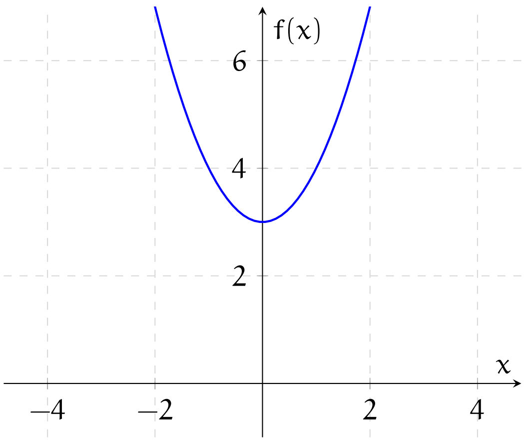
Definition 2.2 (Vertical Transformation).

$$f(x) + k \quad \text{Shift Up}$$

$$f(x) - k \quad \text{Shift Down}$$

Example 2.1 (Vertical Transformation).

$$f(x) = x^2 + 3$$



Note: Key points are $(1, 1), (0, 0), (-1, 1)$.

2.1.2 Horizontal

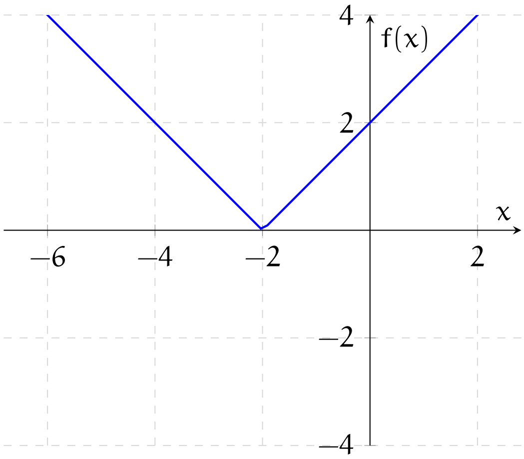
Definition 2.3 (Horizontal Transformation).

$$f(x + h) \quad \text{Shift Left}$$

$$f(x - h) \quad \text{Shift Right}$$

Example 2.2 (Horizontal Transformation).

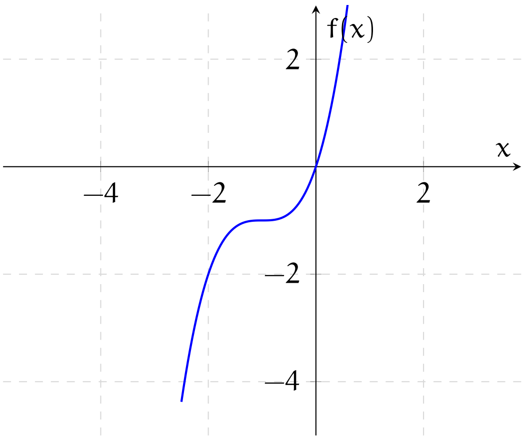
$$f(x) = |x + 2|$$



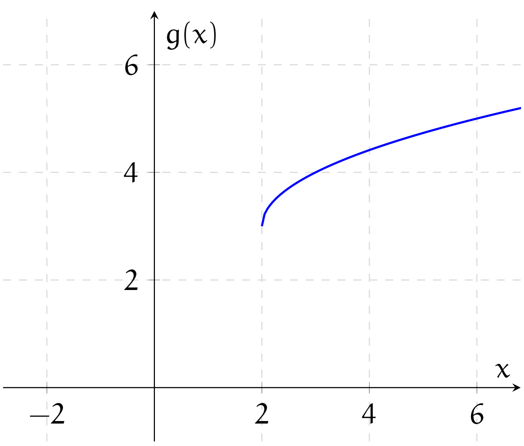
Note: Key points are $(1, 1), (0, 0), (-1, 1)$.

Example 2.3 (Examples).

1.
$$f(x) = (x+1)^3 - 1$$
$$f(x) = (x+1)^3 - 1$$



2.
$$g(x) = \sqrt{x-2} + 3$$



2.1.3 Vertical Stretch/Compression

Definition 2.4 (Vertical Stretch/Compression).

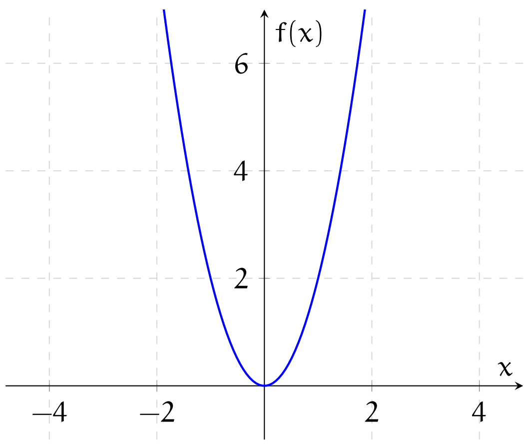
$$y = a\,f(x)$$

$$a > 1 \quad \text{Stretch Narrow}$$

$$a < 1 \quad \text{Compression Wide}$$

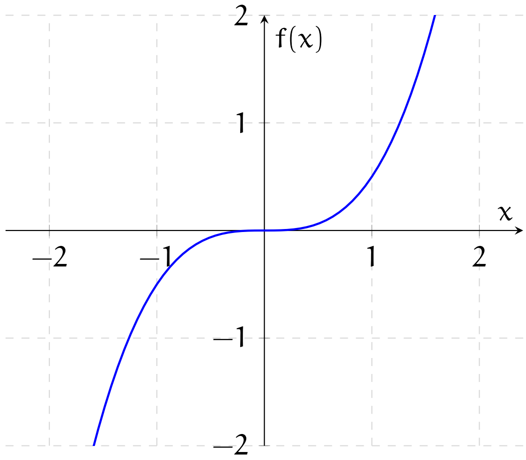
Example 2.4 (Vertical Strectch/Compression).

1. $$f(x) = 2x^2$$



Note: We just multiply our outputs by 2.

2. $$f(x) = \frac{1}{2}x^3$$



2.1.4 Horizontal Stretch/Compression

Definition 2.5 (Horizontal Stretch/Compression).

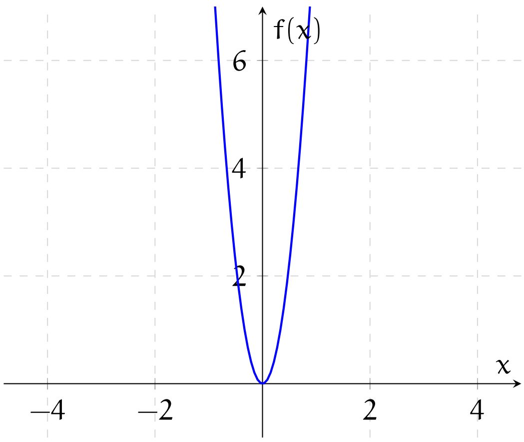
$$y = f(a\,x)$$

$$a > 1 \quad \text{Horizontal Compression}$$
$$a < 1 \quad \text{Horizontal Stretch}$$

Note: Most of the time we can change horizontal stretch/compressions into vertical stretch/compressions respectively.

Example 2.5 (Horizonal Stretch/Compression).

$$f(x) = (3x)^2 \to 9x^2$$



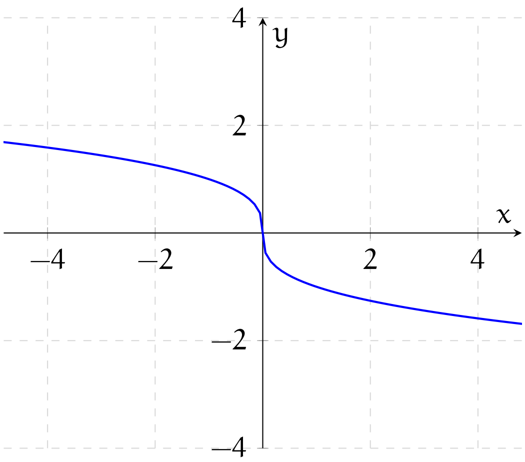
2.1.5 Reflection

Definition 2.6 (Reflection About x Axis).

$$y = -f(x)$$

Example 2.6 (Reflection About x Axis).

$$f(x) = -\sqrt[3]{x}$$

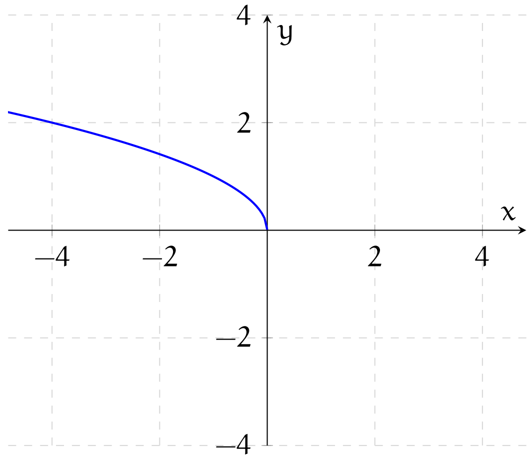


Definition 2.7 (Reflection About y Axis).

$$y = f(-x)$$

Example 2.7 (Reflection About y Axis).

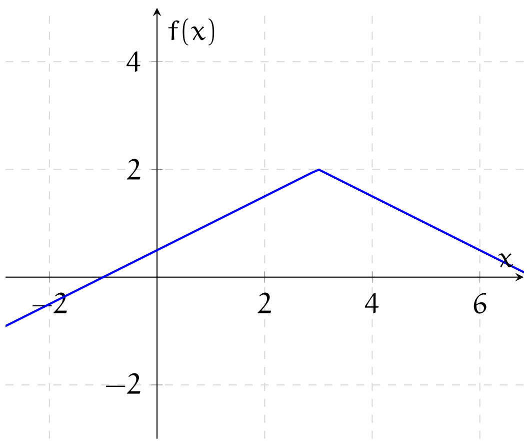
$$f(x) = \sqrt{-x}$$


4
y
2
x
−4
−2
2
4
−2
−4

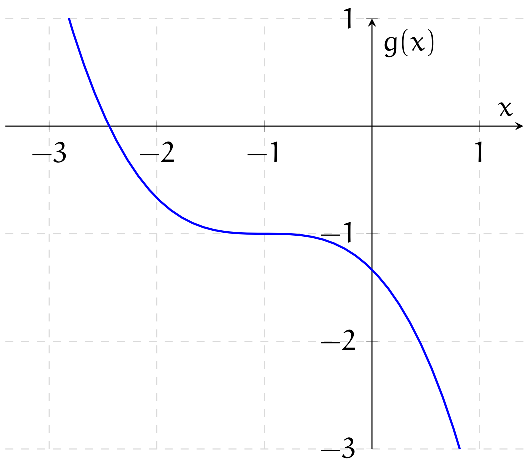
2.2 How to Graph with Transformations

Example 2.8 (Examples).

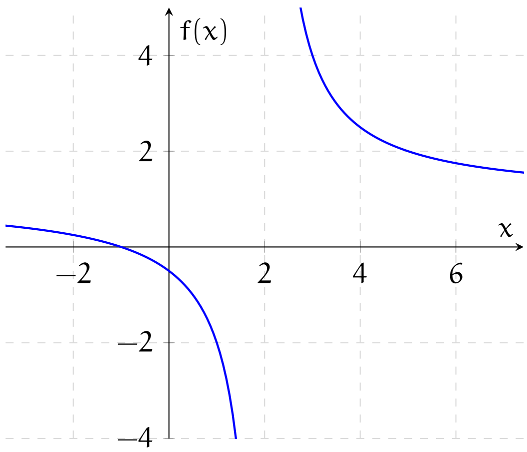
1. $f(x) = -\frac{1}{2}|x-3|+2$
   Key Points $= (1,1)\ (0,0)\ (-1,1)$
   Steps:
   (a) Up 2.
   (b) Right 3.
   (c) Vertical Compression.
   (d) Reflect About x.



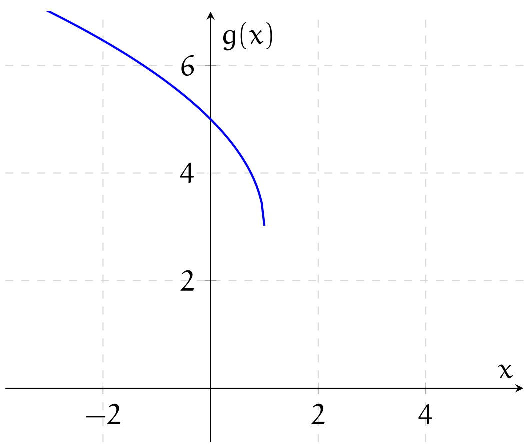
2. $g(x) = -\frac{1}{3}(x+1)^3-1$
   Key Points $= (1,1)\ (0,0)\ (-1,-1)$
   Steps:
   (a) Down 1.
   (b) Left 1.
   (c) Vertical Compression.
   (d) Reflect about x.



3. $f(x) = \frac{3}{x-2} + 1$
   $f(x) = \frac{3}{x-2} + 1 \rightarrow 3.\frac{1}{x-2} + 1$
   Key Points $= (1,1)\ (-1,-1)$
   Steps:
   (a) Up 1.
   (b) Right 2.
   (c) Vertical Stretch.



4. $g(x) = 2\sqrt{1-x} + 3$ $g(x) = 2\sqrt{1-x} + 3 \rightarrow 2\sqrt{-x+1} + 3$
   $g(x) = 2\sqrt{-(x-1)} + 3$
   Key Points $= (1,1)\ (0,0)$
   Steps:
   (a) Up 3.
   (b) Right 1.
   (c) Reflect About x.
   (d) Vertical Stretch.



3 QUADRATICS

3.1 Intro to Solving Quadratics

Definition 3.1 (Standard Form).

$$f(x) = ax^2 + bx + c$$
$$f(x) = 0$$
$$ax^2 + bx + c = 0$$

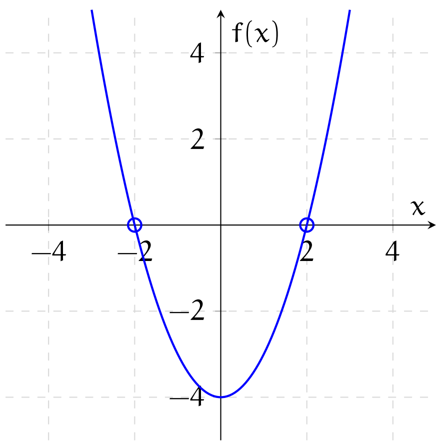
- a, b and c are known numbers (coefficients and constants).
- a cannot be 0, if $a = 0$, the equation becomes linear.

Note: All we really do is just finding the x intercepts.

3.1.1 Cases

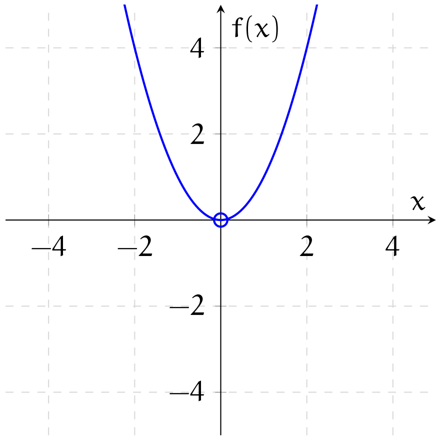
There are a total of 3 cases a quadratic can have.

Example 3.1 (2 x Intercepts).



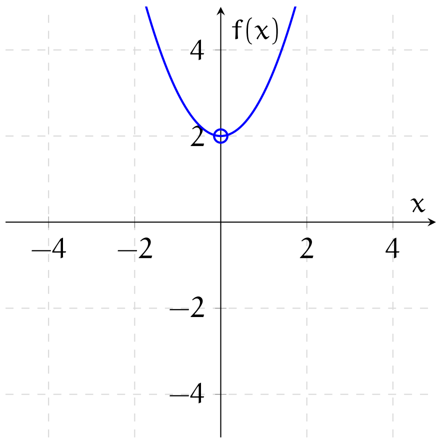
- 2 Solutions.
- 2 Real Solutions.
- 2 Real x Intercepts.

Example 3.2 (1 x Intercept).



- 1 Real Solutions.
- 1 Real x Intercepts.

Example 3.3 (No x Intercept).



- 2 Complex Solutions.
- No x Intercepts.

3.2 The Square Root Method for Quadratics

Definition 3.2 (Square Root Method). A technique used to solve quadratic equations by isolating the squared variable term, then taking the square root of both sides.

Example 3.4 (Square Root Method).

All of these has 2 solutions.

1. $f(x) = x^2 - 18$
   $x^2 - 18 = 0$
   $x^2 = 18$
   $\sqrt{x^2} = \pm\sqrt{18}$
   $x = \pm 3\sqrt{2}$
   $x = 3\sqrt{2} \wedge x = -3\sqrt{2}$

2. $f(x) = 3x^2 - 33$
   $3x^2 - 33 = 0$
   $3x^2 = 33$
   $x^2 = 11$
   $\sqrt{x^2} = \pm\sqrt{11}$
   $x = \pm\sqrt{11}$
   $x = \sqrt{11} \wedge x = -\sqrt{11}$

3. $g(x) = (x+2)^2 - 1$
   $(x+2)^2 - 1 = 0$
   $(x+2)^2 = 1$
   $\sqrt{(x+2)^2} = \pm\sqrt{1}$
   $x + 2 = \pm 1$
   $x = -2 \pm 1$
   $x = -2 - 1 = -1 \wedge x = -2 + 1 = -3$

4. $h(x) = (2x+3)^2 - 32$
   $(2x+3)^2 - 32 = 0$
   $(2x+3)^2 = 32$
   $\sqrt{(2x+3)^2} = \pm\sqrt{32}$
   $2x + 3 = 4\sqrt{2}$
   $2x + 3 = \pm 4\sqrt{2}$
   $2x = -3 \pm 4\sqrt{2}$
   $x = \frac{-3 \pm 4\sqrt{2}}{2}$
   $x = \frac{-3-4\sqrt{2}}{2} \wedge x = \frac{-3+4\sqrt{2}}{2}$

Definition 3.3 (Imaginary Numbers). An imaginary number is any number that can be written as a real number multiplied by the imaginary unit i where $i = \sqrt{-1}$.

Example 3.5 (Special Cases).

1. **1 Solution.**
   $f(x) = (x-7)^2$
   $(x-7)^2 = 0$
   $\sqrt{(x-7)^2} = \pm\sqrt{0}$
   $x - 7 = \pm 0$
   $x = 7 \pm 0$
   $x = 7 - 0 = 7 \wedge x = 7 + 0 = 7$

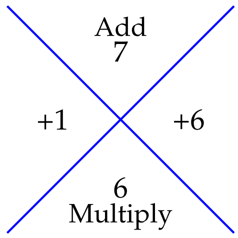
2. **No solution (Imaginary Number).**
$f(x) = (3x-2)^2 + 75$
$(3x-2)^2 + 75 = 0$
$(3x-2)^2 = -75$
$\sqrt{(3x-2)^2} = \pm\sqrt{-75}$
$3x - 2 = \pm 5i\sqrt{3}$
$x = \frac{2 \pm 5i\sqrt{3}}{3}$
$x = \frac{2+5i\sqrt{3}}{3} \wedge x = \frac{2+5i\sqrt{3}}{3}$
Can also be written as
$x = \frac{2}{3} + \frac{5\sqrt{3}}{3}i \wedge x = \frac{2}{3} - \frac{5\sqrt{3}}{3}i$

3.3 Factoring Method for Quadratics

Definition 3.4 (Factoring). Factoring a quadratic is the process of rewriting a standard quadratic expression $ax^2 + bx + c$ as the product of its linear factors, like $(x - p)(x - 1)$.

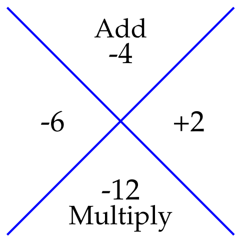
Example 3.6 (Factoring).

1. $f(x) = x^2 + 7x + 6$
   $x^2 + 7x + 6 = 0$



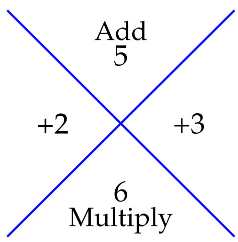
   $x^2 + 1x + 6x + 6 = 0$
   $x(x + 1) + 6(x + 1) = 0 \rightarrow (x + 1)(x + 6) = 0$
   $x + 1 = 0 \wedge x + 6 = 0$
   $x = -1 \wedge x = -6$

2. $g(x) = 3x(x - 4) - 36$
   $3x(x - 4) - 36 = 0$
   $3x^2 - 12x - 36 = 0$
   $3(x^2 - 4x - 12) = 0$

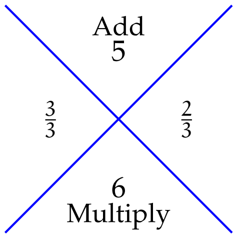


   $3(x^2 - 6x + 2x - 12) = 0$
   $3[x(x - 6)2(x - 6)] = 0$
   $3(x - 6)(x + 2) = 0$
   $x - 6 = 0 \wedge x + 2 = 0$
   $x = 6 \wedge x = -2$

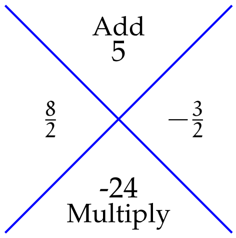
3. $g(x) = 3x^2 + 2 + 5x$
   $3x^2 + 5x + 2 = 0$



$3x^2 + 3x + 2x + 2 = 0$
$3x(x + 1) + 2(x + 1) = 0$
$(x + 1)(3x + 2) = 0$
$x + 1 = 0 \wedge 3x + 2 = 0$
$x = -1 \wedge x = -\frac{2}{3}$
Note: The full Shortcut is (We need to divide the factors by a):



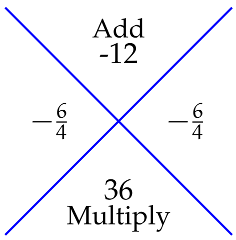
4. $h(x) = -2x^2 - 5x + 12$
   $-2x^2 - 5x + 12 = 0$
   Note: We can change the signs here.
   $2x^2 + 5x - 12 = 0$



$2x^2 + 8x - 3x - 12 = 0$
$2x(x + 4) - 3(x + 4) = 0$
$(x + 4)(2x - 3) = 0$
$x + 4 = 0 \wedge 2x - 3 = 0$
$x = -4 \wedge x = \frac{3}{2}$

Example 3.7 (Special Cases).

1. **1 Solution.**
   $f(x) = -12x + 9 + 4x^2$
   $4x^2 - 12x + 9 = 0$



$(2x - 3)(2x - 3) = 0$
$(2x - 3)^2 = 0$

Note: We can do the square root method here.
$\sqrt{(2x-3)^2} = \pm\sqrt{0}$
$2x-3 = \pm\sqrt{0}$
$(x = \frac{3}{2})(x = \frac{3}{2}) = 0$
$x = \frac{3}{2} \wedge x = \frac{3}{2}$

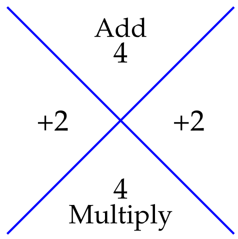
2. **1 Solution.**
$g(x) = x^2 - 5$
$x^2 - 5 = 0$
$x^2 - (\sqrt{5})^2 = 0$
$(x - \sqrt{5})(x - \sqrt{5}) = 0$
$x = \sqrt{5} \wedge x = \sqrt{5}$

3.4 Completing the Square Method for Quadratics

Definition 3.5 (Completing the Square). An algebraic method used to rewrite a quadratic expression from its standard form, $ax^2 + bx + c$, into the vertex form, $a(x - h)^2 + k$. This transformation creates a perfect square binomial.

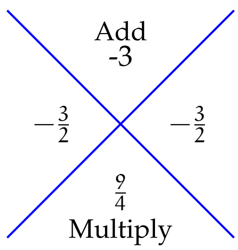
Example 3.8 (Completing the square).

1. $f(x) = x^2 + 4x - 3$
   $0 = x^2 + 4x - 3$
   Note: We're gonna pretend -3 isn't there.
   $0 = x^3 - 4x \qquad -3$



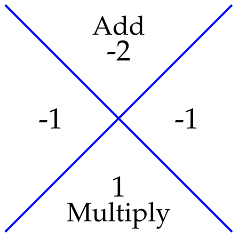
   $0 = (x^2 + 4x + 4) - 4 - 3 \qquad (x + 2)(x + 2) = (x + 2)^2$
   $0 = (x + 2)^2 - 7$
   $(x + 2)^2 = 7$
   $\sqrt{(x + 2)^2} = \pm\sqrt{7}$
   $x + 2 = \pm\sqrt{7}$
   $x = -2 \pm \sqrt{7}$
   $x = -2 + \sqrt{7} \wedge x = -2 - \sqrt{7}$

2. $g(x) = x^2 - 3x + 1$
   $x^2 - 3x + 1 = 0$
   $(x^2 - 3x \qquad) + 1 = 0$



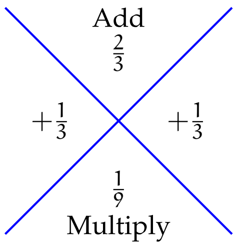
   $(x^2 - 3x + \frac{9}{4}) - \frac{9}{4} + 1 = 0$
   $(x - \frac{3}{2})^2 - \frac{5}{4} = 0$
   $(x - \frac{3}{2})^2 = \frac{5}{4}$
   $\sqrt{(x - \frac{3}{2})^2} = \pm\sqrt{\frac{5}{4}}$
   $x - \frac{3}{2} = \pm\frac{\sqrt{5}}{2}$
   $x = \frac{3}{2} \pm \frac{\sqrt{5}}{2}$
   $x = \frac{3+\sqrt{5}}{2} \wedge x = \frac{3-\sqrt{5}}{2}$

3. $h(x) = 5x^2 - 10x + 2$
   $5x^2 - 10x + 2 = 0$
   $5(x^2 - 2x \qquad) + 2 = 0$



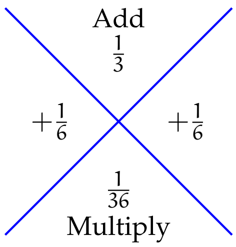
   Note: Remember to subtract 1 by 5 because its multiplied by 5.
   $5(x^2 - 2x + 1) - 5 + 2 = 0$
   $5(x - 1)^2 - 3 = 0$
   $(x - 1)^2 = \frac{3}{5}$
   $\sqrt{(x - 1)^2} = \pm\sqrt{\frac{3}{5}}$
   $x = -1 \pm \sqrt{\frac{3}{5}}$
   $x = -1 - \sqrt{\frac{3}{5}} \wedge x = -1 + \sqrt{\frac{3}{5}}$

4. $g(x) = x^2 + \frac{2}{3}x - \frac{1}{3}$
   $x^2 = \frac{2}{3}x - \frac{1}{3}$
   $(x^2 + \frac{2}{3}x \qquad) - \frac{1}{3}$



   $(x^2 + \frac{2}{3}x + \frac{1}{9}) - \frac{1}{9} - \frac{1}{3} = 0$
   $(x + \frac{1}{3})^2 - \frac{4}{9} = 0$
   $(x + \frac{1}{3})^2 = \frac{4}{9}$
   $\sqrt{(x + \frac{1}{3})^2} = \pm\sqrt{\frac{4}{9}}$
   $x + \frac{1}{3} = \pm\frac{2}{3}$
   $x = -\frac{1}{3} \pm \frac{2}{3}$
   $x = -\frac{1}{3} - \frac{2}{3} \rightarrow \frac{1}{3} \wedge x = -\frac{1}{3} + \frac{2}{3} \rightarrow -1$

5. $f(x) = 3x^2 + x - \frac{1}{2}$
   $3x^2 + x - \frac{1}{2} = 0$
   $3(x^2 + \frac{1}{3}x \qquad) - \frac{1}{2} = 0$



$$
\begin{aligned}
&3(x^2 + \tfrac{1}{3}x + \tfrac{1}{36}) - \tfrac{1}{12} - \tfrac{1}{2} = 0\\
&3(x + \tfrac{1}{6})^2 - \tfrac{7}{12} = 0\\
&3(x + \tfrac{1}{6})^2 = \tfrac{7}{12}\\
&(x + \tfrac{1}{6})^2 = \tfrac{7}{36}\\
&\sqrt{(x + \tfrac{1}{6})^2} = \pm\sqrt{\tfrac{7}{36}}\\
&x + \tfrac{1}{6} = \pm\sqrt{\tfrac{7}{36}}\\
&x = -\tfrac{1}{6} \pm \sqrt{\tfrac{7}{36}}\\
&x = -\tfrac{1}{6} + \sqrt{\tfrac{7}{36}} \wedge x = -\tfrac{1}{6} - \sqrt{\tfrac{7}{36}}
\end{aligned}
$$

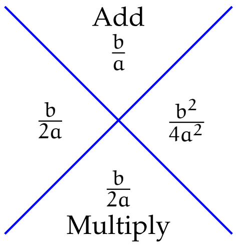
3.5 Proving the Quadratic Formula

Definition 3.6 (Quadratic).

$$f(x) = ax^2 + bx + c = 0$$
$$ax^2 + bx + c = 0$$

3.5.1 Proof

$$\begin{aligned} ax^2 + bx \qquad &+ c = 0 \\ a(x^2 + \frac{b}{a}x \quad) \quad &+ c = 0 \end{aligned}$$



$$a(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}) \quad + c = 0$$
$$a(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}) - \frac{b^2}{4a^2} + c = 0$$
$$a(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}) - \frac{b^2}{4a^2} + \frac{4ac}{4a}$$
$$a(x + \frac{b}{2a})^2 = \frac{b^2}{4a} - \frac{4ac}{4a}$$
$$a(x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a}$$
$$(x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a^2}$$
$$\sqrt{(x + \frac{b}{2a})^2} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$
$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$
$$x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$
$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a} \quad \square$$

3.6 The Quadratic Formula

Definition 3.7 (The Quadratic Formula).

$$\text{For } ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The quadratic formula does the same thing as both the square root method and factoring method. The only difference is that it works all the time.

Note: Completing the square also works all the time but its harder than the quadratic formula.

Example 3.9 (The Quadratic Formula).

1. $f(x) = 3x^2 - 5x + 1$
   $3x^2 - 5x + 1 = 0$
   $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3)(1)}}{2(3)}$
   $x = \frac{5 \pm \sqrt{25-12}}{6} \longrightarrow \frac{5 \pm \sqrt{13}}{6}$
   $x = \frac{5+\sqrt{13}}{6} \wedge x = \frac{5-\sqrt{13}}{6}$
2. $g(x) = 3x(x+2) - 1$
   $3x^2 + 6x - 1 = 0$
   $x = \frac{-(6) \pm \sqrt{(6)^2 - 4(3)(-1)}}{2(3)}$
   $x = \frac{-6 \pm \sqrt{36+12}}{6}$
   $x = \frac{-6 \pm \sqrt{48}}{6} \longrightarrow \frac{-6 \pm 4\sqrt{3}}{6}$
   $x = \frac{-3 \pm 2\sqrt{3}}{3}$
   $x = \frac{-3+2\sqrt{3}}{3} \wedge x = \frac{-3-2\sqrt{3}}{3}$
3. $h(x) = 3x^2 + x - \frac{1}{2}$
   $3x^2 + x - \frac{1}{2} = 0$
   $2(3x^2 + x - \frac{1}{2}) = 0 \cdot 2$
   $6^2 + 2x - 1 = 0$
   $x = \frac{-(2) \pm \sqrt{(2)^2 - 4(6)(-1)}}{2(6)}$
   $x = \frac{-2 \pm \sqrt{4+24}}{12}$
   $x = \frac{-2 \pm \sqrt{28}}{12} \longrightarrow x = \frac{-2 \pm 2\sqrt{7}}{12}$
   $x = \frac{-1 \pm \sqrt{7}}{6}$
   $x = \frac{-1+\sqrt{7}}{6} \wedge x = \frac{-1+\sqrt{7}}{6}$
4. $f(x) = \frac{1}{8}x^2 - \frac{1}{4}x - 2$
   $\frac{1}{8}x^2 - \frac{1}{4}x - 2 = 0$
   $8(\frac{1}{8}x^2 - \frac{1}{4}x - 2 = 0 \cdot 8)$
   $x^2 - 2x - 16 = 0$
   $x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-16)}}{2(1)}$
   $x = \frac{2 \pm \sqrt{4+64}}{2}$
   $x = \frac{2 \pm \sqrt{68}}{2}$
   $x = \frac{2 \pm 2\sqrt{17}}{2} \longrightarrow 1 \pm \sqrt{17}$
   $x = 1 + \sqrt{17} \wedge x = 1 - \sqrt{17}$

Example 3.10 (Special Cases).

1. $f(x) = 9x - 6x + 1$
   $9x - 6x + 1 = 0$
   $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(9)(1)}}{2(9)}$
   $x = \frac{6 \pm \sqrt{36-36}}{18}$
   $x = \frac{6 \pm \sqrt{0}}{18}$
   $\frac{6+0}{18} \wedge = \frac{6-0}{18}$
   $x = \frac{1}{3}$

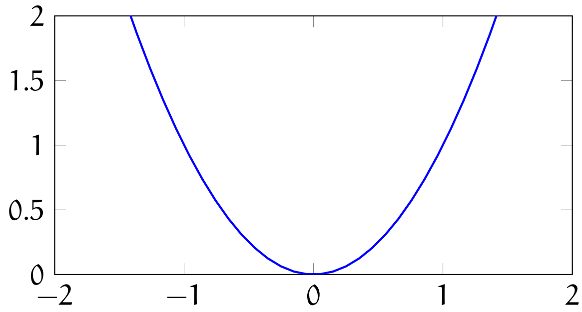
2. $g(x) = -4x^2 - x - 4$
   $-4x^2 - x - 4 = 0$
   Note: We can change all the negatives to positives here.
   $4x^2 + x + 4 = 0$
   $x = \frac{-(1) \pm \sqrt{(1)^2 - 4(4)(4)}}{2(4)}$
   $x = \frac{-1 \pm \sqrt{1-64}}{8}$
   $x = \frac{-1 \pm \sqrt{-63}}{8}$
   $x = \frac{-1 \pm 3i\sqrt{7}}{8}$
   $x = \frac{-1+3i\sqrt{7}}{8} \wedge \frac{-1+3i\sqrt{7}}{8}$

3.7 Graphs of Quadratics

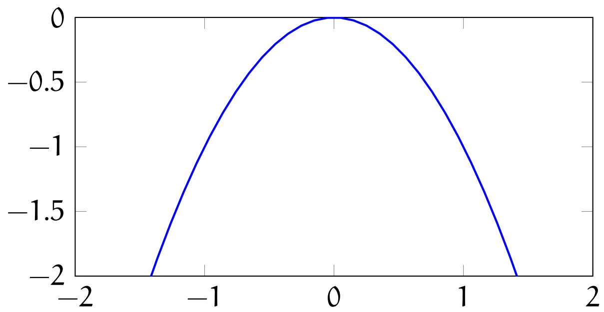
Definition 3.8 (Graphing Quadratics). The process of drawin a U-shaped curve called a *parabola* on a grid based on an equation where the highest power of the variable is 2 ($f(x) = ax^2 + bx + c$).

Note: The graph will always look like if a is.

1. if $a > 0$ it becomes an upward opening parabola.



2. if $a < 0$ it becomes a downward opening parabola.



Definition 3.9 (y Intercept). The point where the graph (a U-shaped curve called a parabola) crosses the vertical y-axis, always occuring where $x = 0$. The y-intercept is simply the constant number c.

$$(0, c)$$

Definition 3.10 (Vertex). The turning point on a parabola's U-shaped graph where the function reaches its maximum or minimum value, and the graph changes direction.

$$\left(\frac{b}{2a}, f(x)\right)$$

Or we can complete the square for the vertex form.

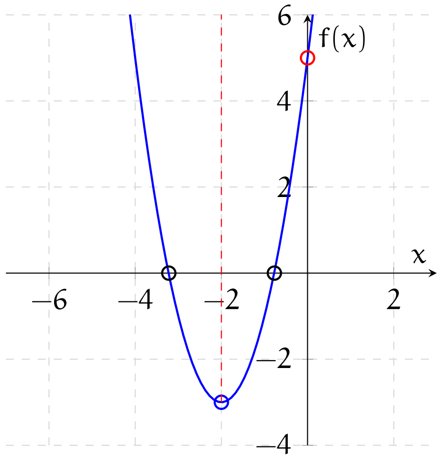
Example 3.11 (Quadratics Graphs).

1. $f(x) = 2x^2 + 8x + 5$

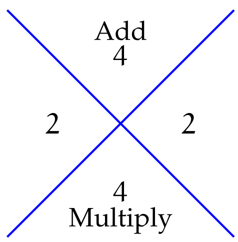
   (a) Vertex:
   $\left(-\frac{(8)}{2(2)}, \quad\right)$
   Note: We plug in -2 into the function.
   $(-2, -3)$

(b) y-intercept:
$(0, 5)$

(c) x-intercept:
$2x^2 + 8x + 5 = 0$
$x = \frac{-(8) \pm \sqrt{(8)^2 - 4(2)(5)}}{2(2)}$
$x = \frac{-8 \pm \sqrt{24}}{4}$
$x \approx -3.22 \wedge x \approx -0.77$



Getting the vertex form:
$f(x) = 2(x^2 + 4x \quad) \quad + 5$



$f(x) = 2(x^2 + 4x + 4) - 8 + 5$
$2(x + 2)^2 - 3$
$f(x) = 2(x + 2)^2 - 3$

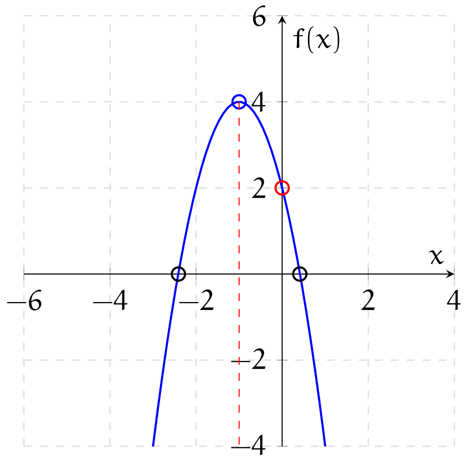
2. $g(x) = -2x^2 - 4x + 2$

(a) Vertex:
$\left(-\frac{(-4)}{2(-2)}, \quad\right)$
$(-1, 4)$

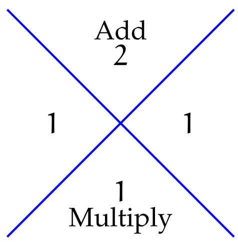
(b) y-intercept:
$(0, 2)$

(c) x-intercept:
$2x^2 - 4x + 2 = 0$
$2(x^2 + 2x - 1) = 0$
$x = \frac{-(2) \pm \sqrt{(2)^2 - 4(1)(-1)}}{2(1)}$

$x = \frac{-2\pm\sqrt{8}}{2}$
$x = \frac{-2\pm2\sqrt{2}}{2}$
$x = -1 \pm \sqrt{2}$
$x \approx 0.41 \wedge x \approx -2.41$



Getting the vertex form:
$g(x) = -2(x^2 + 2x \quad) \quad + 2$



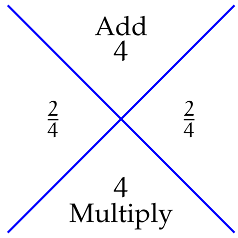
$g(x) = -2(x^2 + 2x + 1) + 2 + 2$
$g(x) = -2(x + 1)^2 + 4$

3. $f(x) = -4x^2 - 4x - 1$

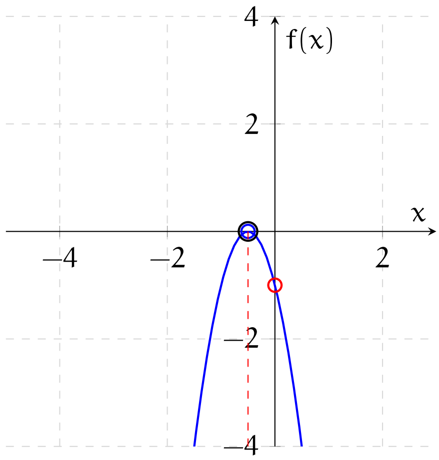
   (a) Vertex:
   $\left(-\frac{(-4)}{2(-4)}, \quad\right)$
   $\left(-\frac{1}{2}, 0\right)$

   (b) y-intercept:
   $(0, -1)$

   (c) x-intercept:
   $-4x^2 - 4x - 1 = 0$
   $4x^2 + 4x + 1 = 0$



$x = -\frac{1}{2}$

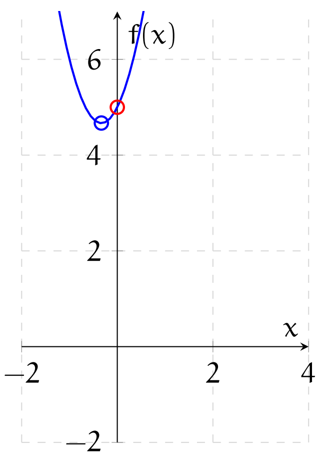


4. $g(x) = 3x^2 + 2x + 5$

   (a) Vertex:
   $\left(-\frac{(2)}{2(3)}, \quad\right)$
   $\left(-\frac{1}{3}, 4\frac{2}{3}\right)$

   (b) y-intercept:
   $(0, 5)$

   (c) x-intercept:
   None.



3.8 Quadratics Inequalities

Definition 3.11 (Quadratics Inequalities). A math sentence with a squared term where the highest power is 2, and it uses signs like $>$, $<$, $\geq$, or $\leq$ instead of a $=$ sign.

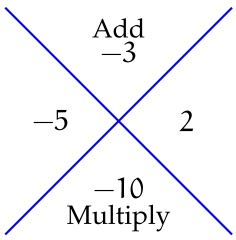
The standard forms is as follows:

$a \neq 0$

- $ax^2 + bx + c > 0$
- $ax^2 + bx + c < 0$
- $ax^2 + bx + c \geq 0$
- $ax^2 + bx + c \leq 0$

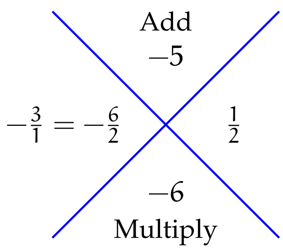
Example 3.12 (Quadratics Inequalities).

1. $x^2 - 3x - 10 \leq 0$
   $x^2 - 3x - 10 = 0$



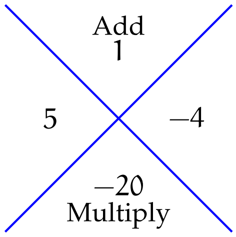
   $(x-5)(x+2) = 0$
   $x - 5 = 0 \wedge x + 2 = 0$
   $x = 5 \wedge x = -2$
   $[-2, 5]$
   $(0)^2 - 3(0) - 10 \leq 0$
   $v(-10 \leq 0) = \top$

2. $2x^2 < 5x + 3$
   Note: We can actually change the inequalities and still get the same answer.
   $2x^2 - 5x - 3 < 0$



   $x = 3 \wedge x = -\frac{1}{2}$
   $(-\frac{1}{2}, 3)$
   $2(0) - 5(0) - 3 < 0$
   $v(-3 < 0) = \top$

3. $x(x+1) > 20$
   $x^2 + x > 20$
   $x^2 + x - 20 > 0$



   $(x+5)(x-4)0$
   $x = -5 \wedge x = 4$
   $(-\infty, -5) \cup (4, \infty)$
   $(0)^2 + (0) - 20 > 0$
   $v(-20 > 0) = \perp$

4. $4x^2 + 9 < 6x$
   $4x^2 - 6x + 9 < 0$
   $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(4)(9)}}{2(4)}$
   $x = \frac{6 \pm \sqrt{36 - 144}}{8}$
   Note: This parabola has **NO** x-axis.
   $4(0)^2 - 6(0) + 9 < 0$
   $v(9 < 0) = \perp$

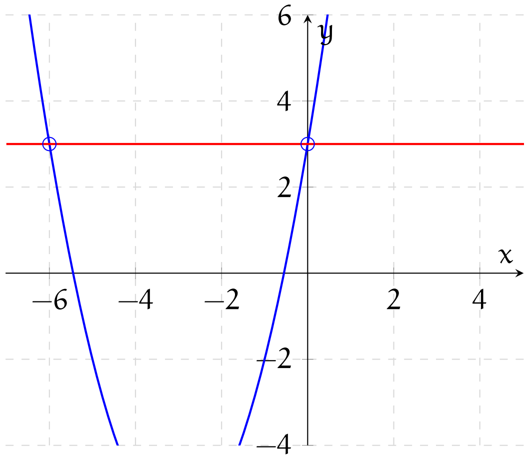
4 Intersection of Functions

Definition 4.1 (Intersection of Functions). The point or set of points where both functions share the exact same input and output values.

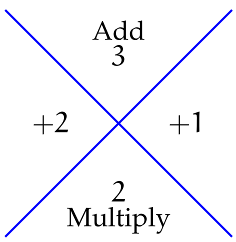
Example 4.1 (Intersection of Functions).

1. $f(x) = x^2 + 6x + 3 \quad g(x) = 3$
   $x^2 + 6x + 3 = 3$
   $x^2 + 6x + 3 - 3 = 0$
   $x^2 + 6x = 0$
   $x(x + 6) = 0$
   $x = 0, x + 6 = 0$
   $x = 0, x = -6$

$$(0, 3), (-6, 3)$$

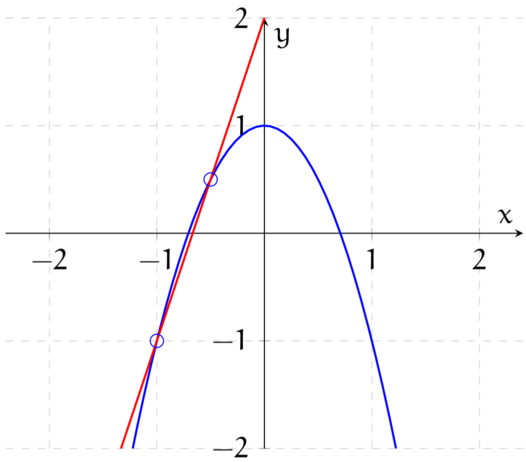


2. $-2x^2 + 1 \quad g(x) = 3x + 2$
   $-2x^2 + 1 = 3x + 2$
   $0 = 2x^2 + 3x - 1 + 2$
   $0 = 2x^2 + 3x + 1$

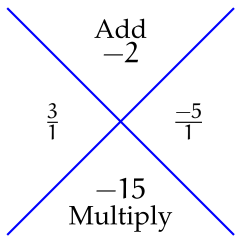


$0 = (x + 1)(2x + 1)$
$x + 1 = 0, 2x + 1 = 0$
$x = -1, x = -\frac{-1}{2}$

$$(-1, -1)(-\frac{1}{2}, \frac{1}{2})$$

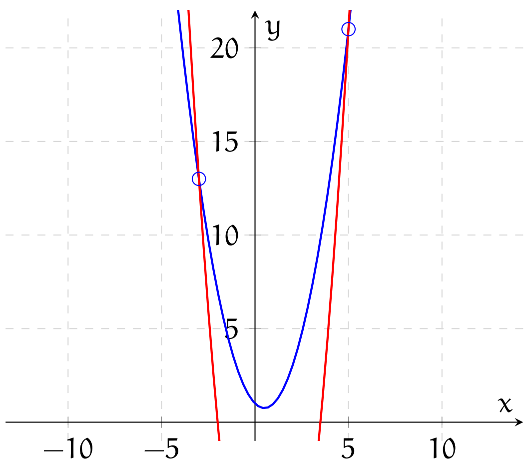


3. $f(x) = x^2 - x + 1 \quad g(x) = 2x^2 - 3x - 14$
$x^2 - x + 1 = 2x^2 - 3x - 14$
$0 = -x^2 + 2x^2 + x - 3x - 1 - 14$
$0 = x^2 - 2x - 15$



$0 = (x+3)(x-5)$
$x + 3 = 0, x - 5 = 0$
$x = -3, x = 5$

$(-3, 13)(5, 21)$



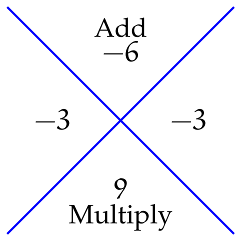
5 Solving Equations by Using Substitutions

Note: also called u-substitution.

Definition 5.1 (u-substitution). Solves equations of quadratic form by rewriting them as $a(g(x))^2 + b(g(x)) + c = 0$ where $u = g(x)$ transforms the equation into the standard quadratic form $au^2 + bu + c = 0$.

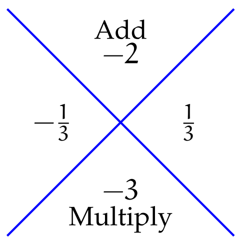
Example 5.1 (Equations).

1. $g(x) = (3x+4)^2 - 6(3x+4) + 9$
   $(3x+4)^2 - 6(3x+4)^1 + 9 = 0$
   $u = 3x + 4$
   $u^2 - 6u + 9 = 0$

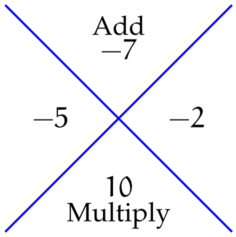
   $(u-3)(u-3) = 0$
   $(u-3)^2 = 0$
   $(u-3) = 0$
   $u = 3 \to 3x + 4 = 3 \to 3x + 4 = 3$
   $3x = -1$
   $x = -\frac{1}{3}$

2. $f(x) = 3x^4 - 2x^2 - 1$
   $3x^4 - 2x^2 - 1 = 0$
   $3(x^2)^2 - 2(x^2) - 1 = 0$
   $u = x^2$
   $3u^2 - 2u - 1 = 0$

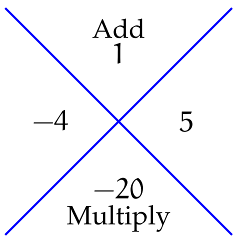
   $u = 1 \wedge u = -\frac{1}{3}$
   $(u-1)(3u+1) = 0$
   $u - 1 = 0 \wedge 3u + 1 = 0$
   $x^2 = 1 \wedge x^2 = -\frac{1}{3}$
   $\sqrt{x^2} = \pm\sqrt{1} \wedge \sqrt{x^2} = \pm\sqrt{-\frac{1}{3}}$
   $(x = 1 \wedge = x = i\sqrt{\frac{1}{3}}) \wedge (x = -1 \wedge = x = -i\sqrt{\frac{1}{3}})$

3. $f(x) = x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10$
   $x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10 = 0$
   $(x^{\frac{1}{3}})^2 - 7(x^{\frac{1}{3}}) + 10 = 0$
   $u = x^{\frac{1}{3}}$
   $u^2 - 7u + 10 = 0$



   $(u-5)(u-2) = 0$
   $u - 5 = 0 \wedge u - 2 = 0$
   $u = 5 \wedge u = 2$
   $x^{\frac{1}{3}} = 5 \wedge x^{\frac{1}{3}} = 2$
   Note: We can do two methods here.
   $\sqrt[3]{x} = 5$
   $(\sqrt[3]{x})^3 = (5)^3$
   $x = 125$
   $x^{\frac{1}{3}} = 2$
   $(x^{\frac{1}{3}})^3 = (2)^3$
   $x = 8$

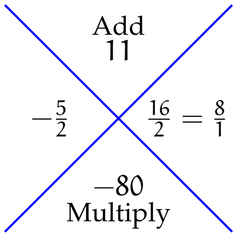
4. $g(x) = x + \sqrt{x} - 20$
   $x + \sqrt[1]{x^2} - 20 = 0$
   $x^1 + x^{\frac{1}{2}} - 20 = 0$
   $(x^{\frac{1}{2}})^2 + (x^{\frac{1}{2}}) - 20 = 0$
   $u = x^{\frac{1}{2}}$
   $u^2 + u - 20 = 0$



   $u = 4 \wedge u = -5$
   $x^{\frac{1}{2}} = 4 \wedge x^{\frac{1}{2}} = -5$
   $\sqrt{x} = 4 \wedge \sqrt{x} = -5$
   $(\sqrt{x})^2 = (4)^2 \wedge (\sqrt{x})^2 = (-5)^2$
   $x = 16 \wedge x = 25$
   Note: $x = 25$ is not a real solution because a square root can't equal a negative.

5. $h(x) = -2y^{-2} - 11y^{-1} + 40$
   $-2y^{-2} - 11y^{-1} + 40 = 0$
   $2y^{-2} + 11y^{-1} - 40 = 0$

$2(y^{-1})^2 + 11(y^{-1}) - 40 = 0$
$u = y^{-1}$
$2u^2 + 11u - 40 = 0$



$(2u - 5)(u + 8) = 0$
$2u - 5 = 0 \wedge u + 8 = 0$
$u = \frac{5}{2} \wedge u = -8$
$y^{-1} = \frac{5}{2} \wedge y^{-1} = -8$
$\frac{1}{y} = \frac{5}{2} \wedge \frac{1}{y} = -8$
$y = \frac{2}{5} \wedge y = -\frac{1}{8}$